

Muthanna K. Kareem

Ph.D. Researcher | Computational Mechanics | Computational Materials Science | Crystal Plasticity | Finite Element Modeling | Mechanics Experimental Testing

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Professional Profiles

Google Scholar: <https://scholar.google.com/citations?user=R8xlv-8AAAAAJ&hl=ar&authuser=2>

LinkedIn: <https://www.linkedin.com/in/muthanna-kareem-99867b71/>

Summary

Mechanical Engineer and Ph.D. Researcher with expertise in computational mechanics, crystal plasticity, finite element modeling, and computational materials science. My research develops predictive computational methods that integrate finite element analysis, image-based modeling, density functional theory, molecular dynamics, and machine learning to understand and optimize the behavior of advanced engineering materials. My work spans biomaterials, structural metals, and energy-storage materials, with multiple peer-reviewed publications and ongoing interdisciplinary collaborations aimed at advancing next-generation materials design.

Research Experties

Computational Mechanics: Finite Element Analysis (FEA), Crystal Plasticity Finite Element Modeling (CPFEM),

Computational Homogenization, Constitutive Modeling

Computational Materials Science: Image-Based Modeling, Representative Volume Elements (RVEs), Multiscale Modeling

Atomistic Modeling: Density Functional Theory (DFT), Molecular Dynamics (MD)

Experimental Characterization: Nanoindentation, μ CT, EBSD, Mechanical Testing

Scientific Computing: Python, MATLAB, Workflow Automation, Machine Learning

Education

University of New Hampshire — Durham, New Hampshire, USA

Ph.D. in Mechanical Engineering (Expected 2028)

|2024–Present | GPA: 3.80/4.00

Research Focus: Computational Mechanics, Crystal Plasticity, Image-Based Finite Element Modeling, Multiscale Modeling, and Computational Materials Science.

Rutgers, The State University of New Jersey — New Brunswick, New Jersey, USA

M.S. in Mechanical Engineering

|2014–2018 | GPA: 3.34/4.00

Thesis: Mechanical Characterization of Thin-Film Oligomeric Silsesquioxane Polymer Coatings on Silicon Using Nanoindentation.

University of Denver, USA

Certificate in English Language

|Jan 2015-Sep 2015|

Al-Furat Al-Awsat Technical University — Iraq

B.S. in Mechanical Engineering (Automobile Engineering)

|2004–2008 | GPA: 3.90/4.00

Top Two

Senior Project: Investigation of Materials Used in the Coating and Remanufacturing of Automotive Components.

Research Experience

Research Methodologies: Crystal Plasticity Finite Element Modeling (CPFEM), Constitutive Modeling, Finite Element Analysis (FEA), Nanoindentation-Based Material Parameter Calibration, Multiscale Computational Modeling.

Crystal Plasticity & Computational Mechanics

- Developed and validated crystal plasticity finite element (CPFEM) models for BCC tantalum and low-carbon steels to investigate microscale deformation mechanisms and anisotropic plastic behavior.
- Integrated nanoindentation experiments with computational simulations to calibrate constitutive parameters and improve predictions of localized plastic deformation.
- Investigated slip-system activity, grain-boundary strengthening, and geometrically necessary dislocation (GND) evolution to better understand structure–property relationships in polycrystalline materials.
- Applied multiscale computational modeling to connect microscale deformation mechanisms with macroscopic mechanical response.

Image-Based Computational Modeling

- Developed a fully automated computational workflow that converts μ CT images of porous materials into finite element-ready representative volume elements (RVEs) for homogenization-based mechanical analysis.
- Integrated image processing, segmentation, mesh generation, and computational homogenization into a unified image-based finite element modeling framework.
- Predicted effective orthotropic mechanical properties of porous biomaterials directly from three-dimensional microstructural images.
- Improved computational efficiency and reproducibility by minimizing manual preprocessing and integrating multiple analysis steps into a single automated workflow.

Multiscale Computational Modeling

- Combined continuum mechanics, finite element analysis, crystal plasticity, and atomistic simulations to investigate material behavior across multiple length scales.
- Developed computational methodologies linking microstructural characteristics with effective mechanical properties for advanced engineering materials.
- Applied computational homogenization techniques to quantify the influence of microstructure on macroscopic material performance.

Computational Materials Science

- Applied density functional theory (DFT), molecular dynamics (MD), and machine learning to investigate advanced energy-storage materials, multifunctional composites, and nanofluids.
- Collaborated on interdisciplinary computational research involving atomistic simulations, data-driven materials design, and predictive modeling of functional materials.
- Contributed to the development of computational approaches for understanding structure–property relationships in emerging engineering materials.

Publications

- 1: Wanni, J., Adkins, C., Rasmussen, R., **Kareem, M. K.**, Clausen, B., Kvit, A., Ma, T., Schreiber, D. K., Yano, K. H., El Atwani, O., Knezevic, M., Bronkhorst, C. A., & Thoma, D. J. (2026). *Deformation of Quenched and Tempered Low-Carbon Steel: Part I—Quasi-2: Static Behavior*. **Metallurgical and Materials Transactions A**
- Alfryyan, N., Alkhafaji, A., Al Bustanji, S., Sharma, K., **Kareem, M. K.**, Almarashi, A., Shaban, M., & Singh, S. (2026). *Chaos and dynamic instabilities of FGM micro-pipes with internal fluid flow via strain gradient theory*. **Scientific Reports**.
- 3: Singh, N. S. S., Idan, M. F., **Kareem, M. K.**, Mahariq, I., Hussein, M. A., Alrawashdeh, A. I., Sabri, L. S., Taner, M., & Salahshour, S. (2026). *Molecular dynamics investigation of structural and thermophysical properties of capric acid–silica nanocomposites under varying initial pressures*. **Chemical Engineering Journal Advances**, 101426.
- 4: Hsu, C.-Y., Al-Assi, G., **Kareem, M. K.**, Taher, G. N., Ghareeb, O. A., Peyghan, A. A., Dauletbaev, A., Siddikov, R., Latipova, M., Mahariq, I., & Imanov, H. (2026). *Design of advanced catalysts for green ammonia production: Activity and mechanism study*. **Inorganic Chemistry Communications**.
- 5: Basem, A., Al Bustanji, S., **Kareem, M. K.**, Singh, P. K., Almunyif, A., Othayq, M. M., Rajab, H., Arora, H., & Shaban, M. (2026). *Nonlinear dynamics and chaotic behavior of curved self-sustaining BNNTs under combined thermal and magnetic fields*. **Scientific Reports**.
- 6: Mohsen, A. M., Fadwi, R. M., **Kareem, M. K.**, Singh, N. S. S., Al-zahy, Y. M. A., Hamd, S. S., Taner, M., & Sahramaneshi, H. (2026). *Nonlinear dynamic transitions and equilibrium topology of a beam under sequential moving masses*. **Scientific Reports**
- 7: Yadav, N. P., Al-Assi, G., **Kareem, M. K.**, Singh, P. K., Peyghan, A. A., Adab, E. H., Mahariq, I., Imanov, H., Zoyidov, A., Ibragimova, S., & Mamadaliyev, K. (2026). *First-principles and molecular dynamics insights into interfacial interactions of CrTe₂/h-BN van der Waals heterostructures for enhanced anchoring in magnesium-ion batteries*. **Journal of Energy Storage**.
- 8: Yang, H., **Kareem, M. K.**, Du, G., Rakhmonov, F., & Alamri, S. (2026). *Machine Learning–Guided Optimization and Design of Bilayer La₂Ti₂O₇/La_{0.7}Sr_{0.3}MnO₃ Composite Absorbers with Enhanced Ku-Band Performance and Thermal–Mechanical Stability*. **Chinese Journal of Physics**.
- 9: Ul Haq, A., Tariq, M., Tajik, H. A., Safi, M. A., **Kareem, M. K.**, Khan, W., & Zia, T. U. H. (2026). *Enhanced photocatalytic degradation of sunset yellow dye using a novel SrFe₁₂O₁₉/PANI/graphene nanoplatelets ternary nanocomposite under visible light and visible light/H₂O₂ irradiation*. **RSC Advances**.
- 10: Ali, A. B. M., Idan, M. F., **Kareem, M. K.**, et al. (2026). *A Predictive Framework for Evaluating the Thermophysical Properties of Multi-Walled Carbon Nanotube Nanofluids Dispersed in a Water–Ethylene Glycol Base Fluid*. **Scientific Reports**.
- 11: Abdallah, M. H., Ali, A. B. M., **Kareem, M. K.**, et al. (2026). *Efficient Z-Scheme NiTiO₃–Zn₃V₂O₈ Photocatalyst for Visible-Light-Driven Antibiotic Degradation and H₂O₂ Production*. **Surfaces and Interfaces**.
- 12: Vini, M. H., Daneshmand, S., Jasim, D. J., Singh, N. S. S., **Kareem, M. K.**, Salahshour, S., & Naseri, H. (2026). *A Novel Method Using Roasting and Leaching for Extracting Vanadium from the Slag Produced by the Electric Arc Furnace*. **International Journal of Refractory Metals and Hard Materials**, 136, 107612.

- 13:** Abuhassan, Q., Al-Assi, G., **Kareem, M. K.**, Chethan, M., Jumanazarove, D., Mohapatra, T. K., Parasuraman, K., Kumar, S., Umarov, A., Saydullaev, B., & Seytnazarov, M. (2026). *A Novel Two-Dimensional Monolayer Tetragonal Boron Carbide as a Promising Anode for Mg-Ion Batteries*. **Journal of Molecular Modeling**, **32**, 272.
- 14:** Jin, Y., Ali, B., **Kareem, M. K.**, et al. (2025). *Precise Forecasting of Shear Stress, Viscosity, and Density for an Aqueous CuO/CaCO₃/SiO₂ Ternary Hybrid Nanofluid Utilizing Artificial Neural Networks*. **Scientific Reports**.
- 15:** **Kareem, M. K.** (2018). *Mechanical Characterization of Thin Film Oligomeric Silsesquioxane Polymer Coated on Silicon Substrate with Nanoindentation Measurements*. **M.S. Thesis, Rutgers, The State University of New Jersey, Piscataway, NJ, USA.**
- 16:** Lei, L., Ryu, J., Marzoki, K., Sullivan, D., **Kareem, M. K.**, et al. (2017). *Morphological Control of Melting Gel Materials by Electrospray Deposition*.

Professional Engineering Experience

Engineering Manager

Iraqi Ministry of Electricity, Iraq | 2010 – 2023

- Managed engineering operations, quality assurance, and maintenance of 11/33 kV electrical transmission and distribution systems.
- Directed materials testing, quality control, and the commissioning of GIS/AIS substations using industry-leading technologies (ABB, Siemens, Schneider Electric, GE).
- Led multidisciplinary engineering teams, infrastructure projects, and technical operations while ensuring compliance with engineering and safety standards.

Quality Control & Safety Engineer

Asswar Babil Company & Al-Jathibya Electrical Company, Iraq | 2008 – 2010

- Conducted quality assurance, testing, and safety inspections for civil and electrical engineering projects.
- Supported infrastructure construction, equipment commissioning, and maintenance activities.

Teaching Experience

Graduate Teaching Assistant

University of New Hampshire | 2024 – 2026

- Supported undergraduate instruction in Materials Science, Statics, Engineering Design, and SolidWorks (CAD).
- Mentored students through laboratory exercises, engineering design projects, and technical problem-solving.
- Facilitated hands-on learning while assisting with course delivery, assessments, and student success.

Graduate Teaching Assistant

Rutgers, The State University of New Jersey | 2016 – 2017

- Assisted in teaching Solid Mechanics, Mechatronics, and Aerospace Engineering.
- Guided students in engineering analysis, laboratory activities, and course assignments while supporting faculty instruction.

Conference Presentations

- 2025- Slip System Activities in BCC Tantalum Revealed via Nanoindentation and Micropillar Compression: Experiments and Crystal Plasticity Simulations
M. K. Kareem, S. Izadpanah, O. Olanrewaju, K. Jacob, S. Pathak, C. A. Bronkhorst, & M. Knezevic
Oral Presentation, TMS 2025 Annual Meeting & Exhibition, Las Vegas, Nevada, USA.
- 2017- Morphological Control of Melting Gel Materials by Electrospray Deposition
L. Lei, J. Ryu, K. Marzoki, D. Sullivan, **M. K. Kareem**, L. Klein, A. Pelegri, J. Singer, G. Rodriguez, & A. Jitianu
Oral Presentation, MRS 2017 Fall Meeting & Exhibit, Boston, Massachusetts, USA.

Honors, Awards & Scholarships

- Graduate Research/Teaching Assistantship (Fully Funded), Department of Mechanical Engineering, University of New Hampshire, 2024–Present.
- CESEM Graduate Fellowship, University of New Hampshire, USA (2025). Competitive fellowship supporting research in computational mechanics and advanced materials.
- HCED Full Scholarship, Higher Committee for Education Development, Government of Iraq (2016–2018). Awarded a fully

funded graduate scholarship to pursue an M.S. in mechanical engineering at Rutgers, the State University of New Jersey – New Brunswick based on academic excellence.

Technical Skills

Computational Modeling & Simulation

- Finite Element Analysis (FEA), Crystal Plasticity Finite Element (CPFE), Elastic–Plastic Self-Consistent (EPSC) Modeling
- Abaqus (UMAT), ANSYS, MSC Marc, COMSOL Multiphysics, Coreform Cubit, ParaView, Dream3D, SolidWorks (CAD), AutoCAD.

Materials Characterization & Experimental Techniques

- Nanoindentation (Hysitron, NanoVantage), Universal Mechanical Testing (Instron), Creep Testing
- EBSD, SEM, AFM, Microstructural Characterization, Thin-Film Mechanical Characterization, Mechanical Testing & Data Analysis

Electrical Testing & Engineering Instrumentation

- Primary & Secondary Injection Testing & Transformer Testing (DC Resistance & Insulation Resistance)
- CPC 100 Testing Systems & High-Voltage Electrical Testing and Diagnostic Equipment.

Programming & Scientific Computing

- Python, MATLAB, Fortran, C/C++, Julia

Engineering Software

Microsoft Office Suite (Word, Excel, PowerPoint, Project)

Adobe Photoshop, PDF Editing Tools.

Languages

Arabic (Native)

English (Fluent)